



Equilibrium study of the Cu–O–SiO₂ system at various oxygen partial pressures



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ABSTRACT

Phase relationships of the Cu–O–SiO₂ system in various boundary conditions (metallic copper saturation, under oxygen partial pressures of 0.01 atm and 0.21 atm) have been experimentally determined by the equilibration/quenching/EPMA technique up to 1773 K. No ternary compounds were observed. An isopleth (Cu₂O–SiO₂) was used to represent the liquidus in equilibrium with air ($P_{O_2} = 0.21$ atm). One simple eutectic reaction with two primary phases (copper oxide and SiO₂) was found. The eutectic point was determined to be 1336 ± 2 K and 0.23 ± 0.01 mol fraction of SiO₂. Equilibrium at copper saturation was achieved in an inert atmosphere of purified argon gas ($P_{O_2} \approx 10^{-5}$ atm) within the temperature range from 1453 K to the critical point of Cu–O system (1620 K). Above 1620 K, experiments were conducted in a N₂ atmosphere containing 1% O₂ by volume ($P_{O_2} = 0.01$ atm). The copper-saturated results were presented on ternary Cu–O–SiO₂ phase diagrams, and the eutectic point was found to be 1456 ± 2 K and 0.08 ± 0.01 mol fraction of SiO₂, 0.60 ± 0.01 mol fraction of Cu. The experimental results have been compared with previous studies and the phase diagrams calculated by MTDATA 5.10 software and its Mtox 8.1 database. The agreement within the temperature range of 1333–1623 K is good. Phase equilibria in the temperature range from $T = (1673–1773)$ K with SiO₂ as the primary phase has been studied systematically for the first time.

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1. Introduction

The Cu–O–SiO₂ system has attracted wide attention during the past decades, due to its wide applications in catalyst [1–3], solar cell [4,5], advanced material [6,7] and pyrometallurgy [8,9]. Accurate thermodynamic information about this system contributes a lot in the optimization of those applications. Several techniques have been applied to study the phase equilibria of the Cu–O–SiO₂ system [10–19].

The first systematic study of the Cu–O–SiO₂ system in equilibrium with air was by Berezhnoi et al. [10], using the equilibration/quenching/optical microscopy, X-ray and chemical analysis techniques. From the estimated liquid contents of the fused Cu₂O–SiO₂ mixtures, they constructed the first binary phase diagram of Cu₂O–SiO₂, with a simple eutectic point at 1333 ± 10 K and 0.172 mol fraction of SiO₂. Ust'yantsev et al. [11] reinvestigated the phase equilibria of the system in air by equilibration/quenching/X-ray diffraction techniques, the eutectic temperature agrees well with Berezhnoi's [10]. The liquidus information was

not reported in their studies [10,11], which makes the data useless in thermodynamic assessments. Gadalla et al. [12] used a thermobalance to measure the dissociation paths of several CuO–SiO₂ mixtures in different oxygen partial pressures (0.21 atm, 0.50 atm, 1.00 atm), in a study of phase relationships in the CuO–Cu₂O–SiO₂ system. Gadalla's et al. [12] work gave a diagram almost identical with Berezhnoi's [10] phase diagram, but with the eutectic point at 1343 K with 0.152 mol fraction of SiO₂. This study provided valuable information on low temperature phase relationships of the Cu–O–SiO₂ system. Kuxmann and Kurre [13] investigated the effect of silica saturation on the miscibility gap of Cu–Cu₂O system by means of a direct equilibration/sampling technique in temperature range $T = (1459–1683)$ K, and the eutectic point of copper saturated system was reported to be 1459 K and 0.224 mol fraction of SiO₂. Landolt [14] studied the phase equilibria using a gas equilibration technique along several oxygen isobars ($P_{O_2} = 0.001–0.96$ atm) at 1473 K, and the liquid copper + liquid oxide + silica equilibrium between 1473 K and 1700 K was studied in quartz ampoules. The liquidus temperatures reported by Landolt [14] are consistent with Kuxmann and Kurre's [13] results. Luraschi and Elliott [15] summarized the previous studies of this system within the temperature range from 1323 K

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Table 1
Mass fraction purity and the sources of materials used in the present study.

Chemical	Mass fraction purity [*]	Supplier
Cu cathode	0.99996	Boliden Harjavalta (Finland)
Cu powder	0.9996	Alfa Aesar (Germany)
SiO ₂ powder	0.9999	Umicore (Belgium)
Cu ₂ O	0.9999 ^{**}	Synthesized in this work
Argon gas	0.99999	AGA (Finland)
1% O ₂ –99% N ₂ gas	0.9999	AGA (Finland)
Quartz Crucible	0.9998	OM Lasilaite Oy (Finland)
Platinum wire	0.9999	Johnson-Matthey Noble Metals (UK)
Platinum foil	0.9999	Johnson-Matthey Noble Metals (UK)

^{*} Stated by the supplier.

^{**} Derived from EDS results.

to 1673 K, and published a review of the phase relations of the Cu–O–SiO₂ system. Peddada and Gaskell [16] measured the activities of Cu and Cu₂O in the systems of Cu–O and Cu–O–SiO₂ at 1573 K by establishing the equilibrium ($2\text{Cu} + 1/2\text{O}_2 = \text{Cu}_2\text{O}$) between the melts, atmospheric air, and a small piece of platinum wire. The activity of Cu₂O along with air isobar in the Cu–O–SiO₂ system decreases from unity to 0.64 at the solubility limit of $X_{\text{SiO}_2} = 0.13$. Rannikko [17] made a Calphad-type critical assessment of the Cu–O–SiO₂ system, and provided a consistent thermodynamic description of the system over a large temperature and concentration interval. Hidayat et al. [18] applied an equilibration/quenching/EPMA technique to phase equilibria of the Cu–O–Si system. In their study, the eutectic temperature of Cu–O–Si system in air was evaluated to be between 1327 K and 1343 K, and the eutectic point in copper saturated systems was found to be between 1461 K and 1464 K, and approximately 0.227 mol fraction of SiO₂. By employing Factsage software, Hidayat and Jak [19] reassessed the Cu₂O–SiO₂ phase diagram in equilibrium with copper, using modified Quasi-chemical and Bragg-Williams models for the description of the liquid oxide phase. A set of self-consistent model parameters was obtained to reproduce the available experimental data [13,14,18].

In the present study, the phase equilibria of Cu–O–SiO₂ system in various oxygen partial pressures have been investigated using the equilibration/quenching/EPMA technique up to 1773 K. The technique used has been proven to be efficient in acquiring high temperature equilibrium information for geology [20,21] and metallurgy [22]. The technique involved preparation of a synthetic pellet of a predetermined composition. The pellet placed in a suitable container was equilibrated at a target temperature for certain period of time, followed by rapid quenching. The silica-containing liquid converts to glass so that the phase assemblages at high temperatures were frozen instantly. The compositions of equilibrated phases were measured by an Electron-Probe X-ray Micro-analyser (EPMA). The statistical analysis technique enables accurate information to be obtained with estimated experimental uncertainties.

2. Material and methods

2.1. Preparation of initial mixtures

High-purity Cu cathode, Cu powder and SiO₂ powder were employed as starting materials, see Table 1. Cu₂O was prepared by oxidizing an A-grade copper cathode, in air at 1303 K for 144 h [23]. The composition of the oxide was examined by a LEO 1450 (Carl Zeiss Microscopy GmbH, Jena, Germany) scanning electron microscope linked with an Inca X-Sight 7366 Energy EDS analyzer (Oxford Instruments plc, Abingdon, Oxfordshire, UK). The standards used in the EDS analysis were natural minerals (cuprite (Cu₂O) for Cu, L α ; quartz for O, K α). Only Cu and O were detected, and the purity of the ‘Cu₂O’ was included in Table 1. In the equilibrium experiments, the raw materials were dried at 383 K, weighed in desired proportions, mixed thoroughly in an agate mortar, and pelletized at 5 MPa in a 5 mm-diameter pressing tool. Total weight of the pellet was around 0.2 g. The small size of the pellet was an advantage in approaching the equilibria and quenching the molten phases. The pellet was equilibrated at a fixed temperature for a certain period of time, and then quenched in an ice-cold water bath (273 K). Purified argon gas and 1% O₂–99% N₂ mixture were used as flushing gases in the experiments. The purities of the gases are presented in Table 1. Platinum wire and foil, which were inert in

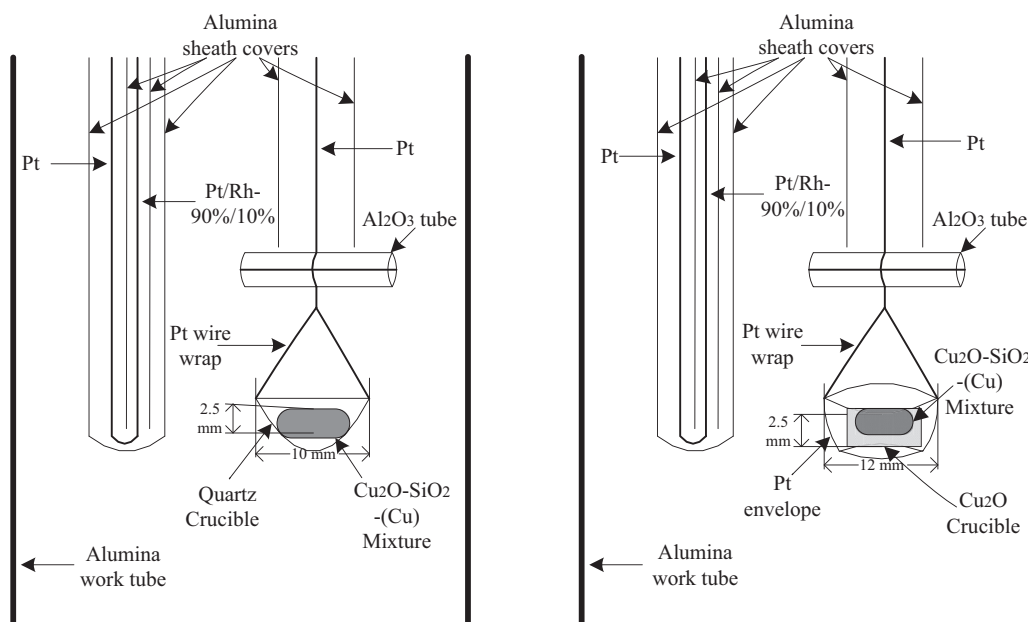


Fig. 1. Schematics of the specimen, container and suspension techniques.

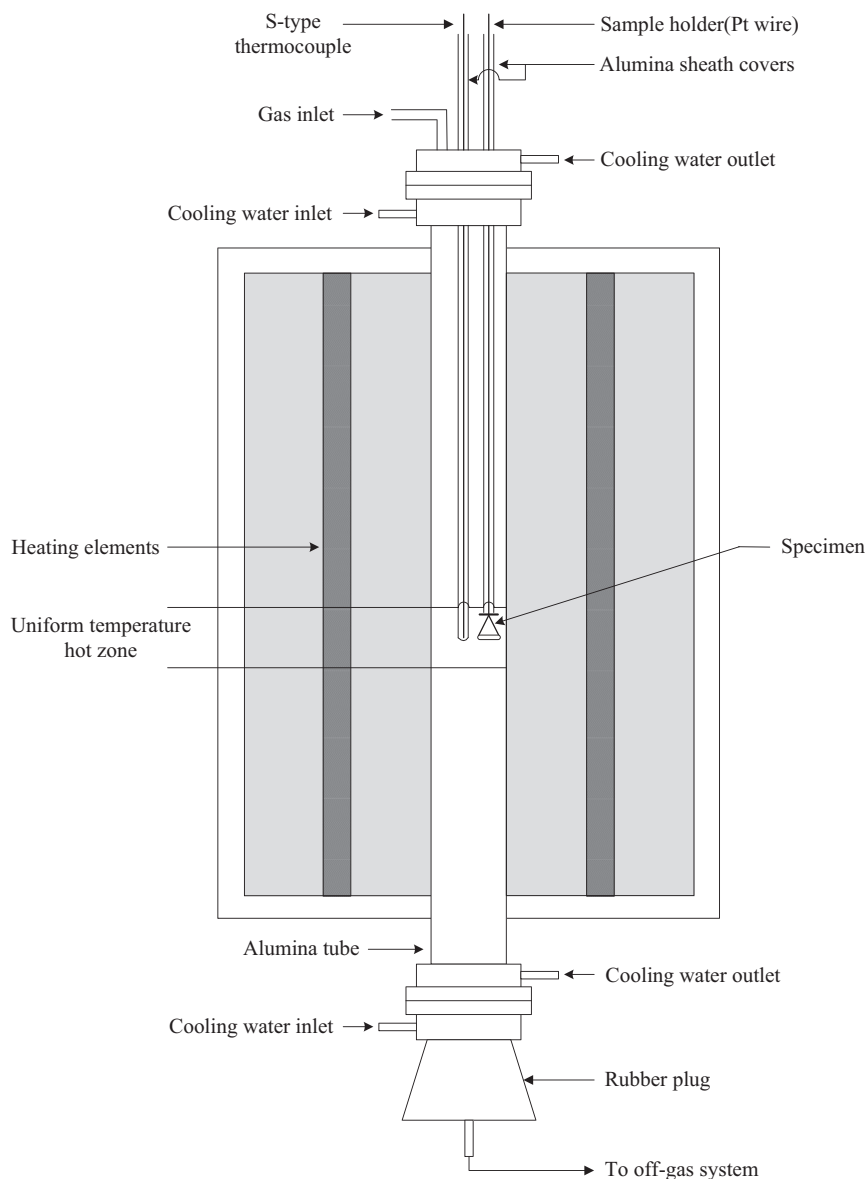


Fig. 2. A schematic vertical section of the furnace and its auxiliaries.

experimental conditions of this study, were used as the suspending and holding wrap to preserve the purity of the system.

2.2. Equilibration technique

Quartz crucibles ($\varnothing$ -10 mm, in Table 1) were utilized in the experiments with SiO_2 as primary phase (Fig. 1 left), which had been proved to be the suitable container in the former study [24]. The experiments with Cu_2O as primary phase were conducted in self-produced Cu_2O crucibles. A small piece of metallic copper cathode was shaped to a crucible, oxidized in air at 1333 K for 12 h [23], and quenched in air to produce the Cu_2O crucible. The pellet was placed in the Cu_2O crucible and equilibrated in a wrap of 0.5-mm platinum foil (Fig. 1 right).

The equilibrium experiments were carried out in a vertical tube furnace (LTF 16/–/450, Lenton, UK) with silicon carbide heating elements (Fig. 2). Special attention was paid to the accuracy of the equilibration temperature. Uniform temperature hot zone of the furnace was first confirmed by measuring the thermal profiles at several temperatures from 1273 K to 1873 K. The temperature of the hot

zone was determined with a calibrated S-type thermocouple (Johnson-Matthey Noble Metals, UK), which was connected to a Keithley 2010 DMM multimeter. The S-type thermocouple was calibrated with metallic copper, zinc and pure water. The collected phase transition temperatures (solid $\rightarrow$ liquid) of copper, zinc and water were 1356.76 ± 0.20 K, 692.40 ± 0.10 K and 273.32 ± 0.10 K, respectively, which were consistent with the literature data [8,25,26]. The working thermocouple in a recrystallized alumina sheath was positioned right next to the specimen. The cold junction compensation was performed by a Pt100 resistance thermometer (SKS-group, Finland; tolerance class B 1/10), connected to a Keithley 2000 DMM multimeter. The overall temperature accuracy was estimated to be ± 2 K for the entire experimental temperature range. The temperature data during the experiments were collected with a National Instruments LabVIEW temperature logging program.

The specimen hooked to a pure platinum wire (0.5 mm diameter) was introduced into the furnace from the bottom of the working tube and suspended in the hot zone. Specimen was kept in the hot zone at the target temperature for the required equilibration time. Particular attention was paid to ensuring the achievement

Table 2Experimentally determined phase compositions for the constrained Cu₂O–SiO₂ system equilibrated with air ($P_{\text{tot}} = 1$ atm, $P_{\text{O}_2} = 0.21$ atm).

No	T°/K	Time/h	Phase	EPMA (mass fraction)**			Recalculated composition (mole fraction)			Standard deviation μ *** (mole fraction)
				O	Cu	Si	Cu	SiO ₂	O	
<i>Cu₂O primary-phase field</i>										
1	1373	12	Liquid Oxide	0.1425	0.8406	0.0169	0.6143	0.0280	0.3577	0.0016
			Copper Oxide	0.1120	0.8878	0.0002	0.6665	0.0003	0.3332	0.0004
<i>Sub-eutectic equilibrium</i>										
2	1333	80	Copper Oxide	0.1121	0.8876	0.0003	0.6663	0.0005	0.3332	0.0003
			Tridymite	0.5290	0.0075	0.4635	0.0071	0.9894	0.0035	0.0006
<i>SiO₂ primary-phase field</i>										
3	1338	60	Liquid Oxide	0.1786	0.7701	0.0513	0.5649	0.0851	0.3500	0.0011
			Tridymite	0.5267	0.0124	0.4609	0.0117	0.9823	0.0060	0.0015
4	1373	36	Liquid Oxide	0.1806	0.7612	0.0582	0.5652	0.0978	0.3370	0.0017
			Tridymite	0.5269	0.0119	0.4612	0.0112	0.9833	0.0055	0.0004
5	1423	24	Liquid Oxide	0.1817	0.7517	0.0666	0.5683	0.1139	0.3178	0.0002
			Tridymite	0.5286	0.0085	0.4629	0.0080	0.9874	0.0046	0.0004
6	1473	24	Liquid Oxide	0.1841	0.7463	0.0696	0.5654	0.1193	0.3153	0.0017
			Tridymite	0.5278	0.0101	0.4621	0.0095	0.9855	0.0050	0.0009
7	1523	24	Liquid Oxide	0.1845	0.7451	0.0704	0.5651	0.1208	0.3141	0.0015
			Tridymite	0.5266	0.0127	0.4607	0.0120	0.9815	0.0065	0.0011
8	1573	24	Liquid Oxide	0.2015	0.7124	0.0861	0.5406	0.1478	0.3116	0.0016
			Tridymite	0.5258	0.0142	0.4600	0.0134	0.9801	0.0065	0.0009
9	1623	24	Liquid Oxide	0.2144	0.6918	0.0938	0.5197	0.1594	0.3209	0.0014
			Tridymite	0.5290	0.0079	0.4631	0.0075	0.9873	0.0052	0.0013
10	1673	12	Liquid Oxide	0.2326	0.6430	0.1244	0.5002	0.2190	0.2808	0.0009
			Tridymite	0.5275	0.0107	0.4618	0.0101	0.9847	0.0052	0.0003
11	1723	12	Liquid Oxide	0.2623	0.5867	0.1510	0.4559	0.2655	0.2786	0.0015
			Tridymite	0.5274	0.0108	0.4618	0.0102	0.9850	0.0048	0.0009
12	1773	12	Liquid Oxide	0.2873	0.5265	0.1862	0.4225	0.3380	0.2395	0.0005
			Cristobalite	0.5261	0.0138	0.4601	0.0130	0.9798	0.0072	0.0009

* Standard uncertainty: $\mu(P_{\text{tot}}) = \pm 5\%$; $\mu(P_{\text{O}_2}) = \pm 5\%$; $\mu(T) = \pm 1$ K.** Average standard uncertainty for EPMA results (in total 240 data points): $\mu(\text{EPMA}) = 0.0008$ mass fraction.

*** Standard deviation of the mean for the recalculated composition of each phase, calculated from 10 individual points.

of equilibrium, by approaching the final equilibrium compositions with variation of the equilibrium time. Achievement of equilibria was determined by measuring the composition of the liquid phase after different annealing times. Three dedicated series of experiments at different temperatures (1373 K, 1573 K and 1773 K) were conducted in air to determine the proper equilibration time. The results indicated that homogenous phases could be achieved within 12 h at 1373 K, and the suitable equilibration period at 1573 K and 1773 K was 8 h and 4 h, respectively. The other three focused time series of experiments at different temperatures (1473 K, 1573 K and 1773 K) were conducted in argon to determine the proper equilibration time for the low oxygen partial pressure experiments. Four hours was sufficient for the experiments conducted below 1573 K, and 0.5 h was found to be enough for the samples at 1773 K.

According to the Gibbs phase rule in isobaric conditions ($F = C - P + 1$, where F is the number of degrees of freedom or independent variables, P is the number of phases, and C is the number of components.), when the system is equilibrated with air ($P_{\text{O}_2} = 0.21$ atm) it has either one or zero degree of freedom. When the molten phase is equilibrated with a solid phase (Cu₂O or SiO₂) at a fixed temperature, the composition of the liquid oxide phase is fully defined. In the copper-saturated Cu–O–SiO₂ systems, in isothermal and isobaric conditions the three-phase equilibria (liquid copper, liquid oxide, and a solid phase (Cu₂O or SiO₂)) have zero degrees of freedom. Thus they were equilibrated in pure argon gas. Above the critical point of Cu–O system (1620 ± 3 K) [17,27,28], liquid copper will disappear because of the miscibility gap. The system will get one more degree of freedom, and therefore e.g. oxygen partial pressure should be fixed to fully define the system above 1620 K.

The experiments equilibrated in air were conducted with the bottom and the top of the reaction tube opened to the ambient

atmosphere ($P_{\text{O}_2} = 0.21$ atm). For metallic copper saturated systems, an inert high-purity argon gas was continuously flushed through the gas tight system under 1620 K. 1% O₂–99% N₂ mixture ($P_{\text{O}_2} = 0.01$ atm) was utilized as gaseous atmosphere for the experiments equilibrated between 1620 K and 1773 K.

The specimen was introduced into the working tube by a platinum wire, and hanged at the bottom. The working tube was sealed with a rubber plug after introduction of the specimen. It was pulled up to the hot zone after 30 min's flushing. Once the equilibration time was reached, the bottom of the tube was immersed into a bottle of water–ice mixture. The specimen was then released rapidly and quenched into the ice-cold water bath. Afterward it was dried and mounted in an epoxy resin. A polished cross section of the mounted specimen was prepared using conventional metallographic grinding and polishing techniques.

2.3. Analysis

All samples were carbon-coated with a Leica EM SCD050 Coater (supplied by Leica Mikrosystems GmbH, Vienna, Austria). The microstructure was examined by a LEO 1450 (Carl Zeiss Microscopy GmbH, Jena, Germany) scanning electron microscope. The compositions of the equilibrated phases were measured by an Electron-Probe X-ray Microanalyser (EPMA). The EPMA analysis was performed by a CAMECA SX100 instrument (Cameca SAS, France), which was equipped with five wavelength dispersive spectrometers. An accelerating voltage of 15 kV, beam current of 40 nA, and spot size of 1–100 μm (diameter) were utilized in the analysis. The standards used in the measurements were natural minerals (quartz for Si: K α ; metallic copper for Cu: K α ; hematite for O: K α). All standards were purchased from Astimex Scientific Ltd, Canada. The raw measurement results were corrected using

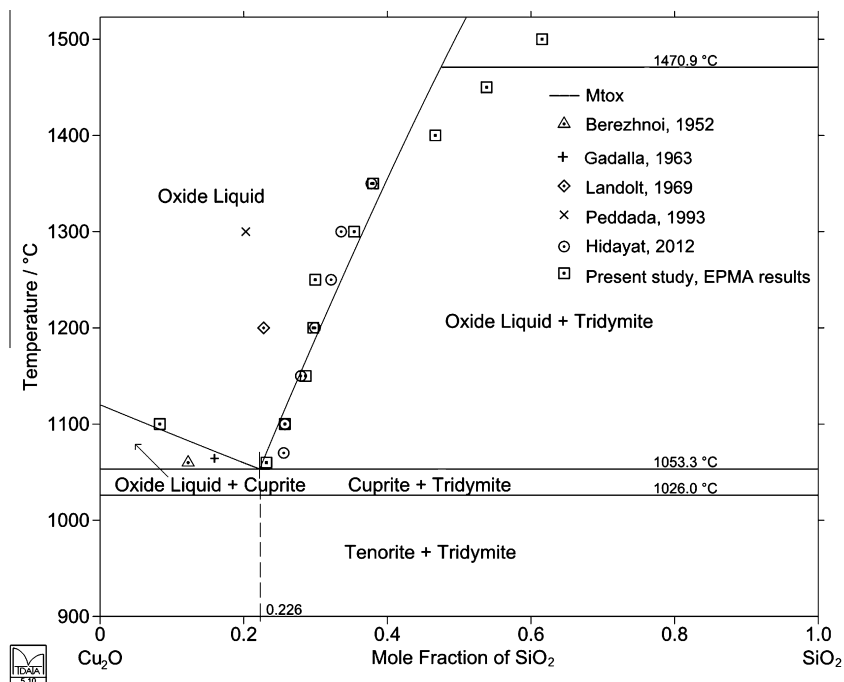


Fig. 3. ‘Cu₂O’-SiO₂ isopleth of the ternary Cu-O-SiO₂ system (in $P_{O_2} = 0.21$ atm), with the experimental points (derived from the mole ratio between Si and Cu in the oxide liquid phase) of the various authors superimposed; the solid lines were calculated using MTDATA 5.10 software and Mtox 8.1 database.

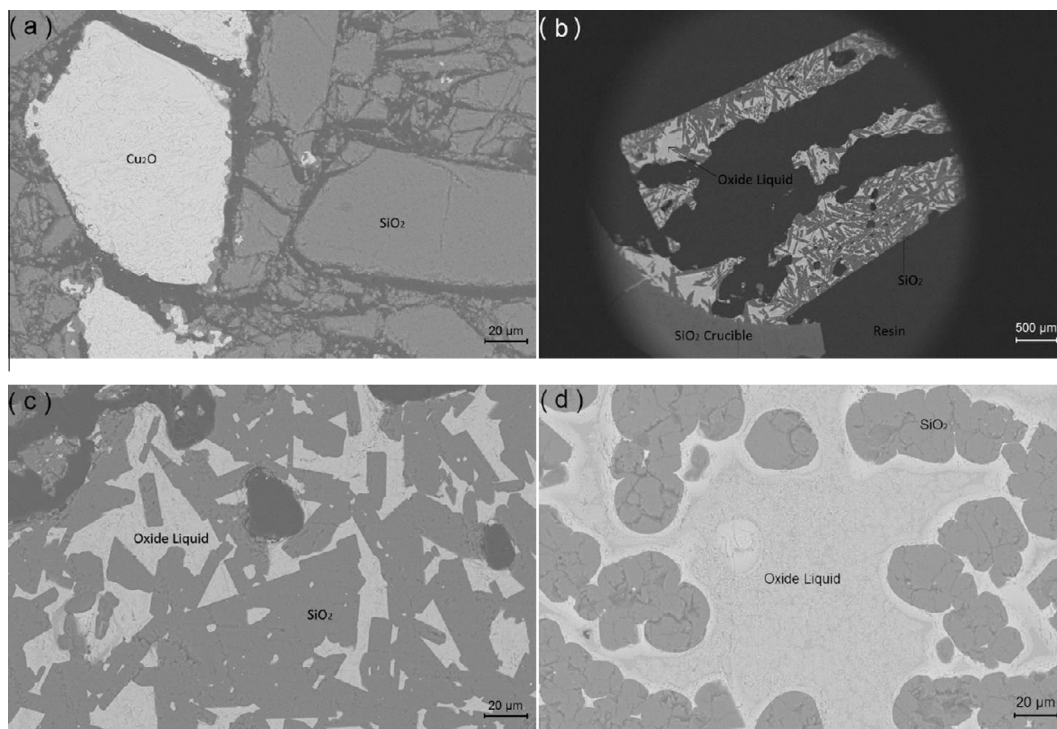


Fig. 4. Backscattered SEM images of selected quenched Cu₂O-SiO₂ samples of this study in air: (a) Table 2, #02; (b) and (c) Table 2, #03 and (d) Table 2, #12.

the PAP on-line matrix correction program [29], supplied by the equipment manufacturer.

3. Results and discussion

3.1. Cu-O-SiO₂ system equilibrated in $P_{O_2} = 0.21$ atm

For the temperature range of interest, two homogeneous phases were observed in each specimen, and no concentration gradients

were detected within them. The compositions of the phases in equilibrium were carefully measured with EPMA. 10 independent measuring points from each phase were analyzed in each sample, taken from its different well-quenched parts, to quantify the chemical composition. The obtained results are listed in Table 2. For a better understanding and graphical comparisons, the chemical compositions have been recalculated based on the measured elements (Cu, O and Si). Cu, O and SiO₂ were considered as the terminal phases for presentation purpose only. The standard deviation of

Table 3Experimentally determined phase compositions of the Cu–O–SiO₂ system. (No. 01 – No.08 at copper saturation, $P_{\text{tot}}^* = 1$ atm; No. 09 – No.12 at $P_{\text{tot}}^* = 1$ atm, $P_{\text{O}_2}^* = 0.01$ atm).

No	T°/K	Time /h	Phase	EPMA (mass fraction)**			Recalculated composition (mole fraction)			Standard deviation μ^{***} (mole fraction)
				O	Cu	Si	Cu	SiO ₂	O	
<i>Three-phase equilibrium of Cu, liquid oxide and Cu₂O</i>										
1	1473	8	Liquid Oxide	0.1473	0.8230	0.0297	0.6138	0.0501	0.3361	0.0018
			Copper Oxide	0.1160	0.8840	0.0000	0.6574	0.0000	0.3426	0.0014
2	1458	8	Liquid Oxide	0.1574	0.7940	0.0486	0.6065	0.0840	0.3095	0.0017
			Copper Oxide	0.1137	0.8863	0.0000	0.6625	0.0000	0.3375	0.0013
<i>Three-phase equilibrium of Cu, liquid oxide and SiO₂</i>										
3	1453	24	Liquid Oxide	0.1150	0.8850	0.0000	0.6596	0.0000	0.3404	0.0010
			Tridymite	0.5326	0.0000	0.4674	0.0000	1.0000	0.0000	0.0000
4	1458	8	Liquid Oxide	0.1597	0.7908	0.0495	0.6022	0.0853	0.3125	0.0015
			Tridymite	0.5326	0.0000	0.4674	0.0000	1.0000	0.0000	0.0000
5	1473	4	Liquid Oxide	0.1617	0.7836	0.0547	0.6018	0.0950	0.3032	0.0019
			Tridymite	0.5284	0.0080	0.4636	0.0076	0.9916	0.0008	0.0011
6	1523	4	Liquid Oxide	0.1727	0.7672	0.0601	0.5825	0.1032	0.3143	0.0008
			Tridymite	0.5304	0.0086	0.4610	0.0080	0.9728	0.0192	0.0021
7	1573	4	Liquid Oxide	0.1688	0.7709	0.0603	0.5908	0.1045	0.3047	0.0022
			Tridymite	0.5298	0.0066	0.4636	0.0062	0.9877	0.0061	0.0014
8	1603	4	Liquid Oxide	0.1690	0.7702	0.0608	0.5907	0.1055	0.3038	0.0006
			Tridymite	0.5298	0.0061	0.4641	0.0057	0.9903	0.0040	0.0009
9	1623	2	Liquid Oxide	0.1817	0.7431	0.0752	0.5740	0.1314	0.2946	0.0019
			Tridymite	0.5280	0.0108	0.4612	0.0101	0.9803	0.0096	0.0015
10	1673	2	Liquid Oxide	0.2038	0.6986	0.0976	0.5427	0.1715	0.2858	0.0017
			Tridymite	0.5284	0.0118	0.4598	0.0110	0.9721	0.0169	0.0008
11	1723	1	Liquid Oxide	0.2133	0.6850	0.1017	0.5261	0.1767	0.2972	0.0024
			Tridymite	0.5296	0.0137	0.4567	0.0127	0.9533	0.0340	0.0014
12	1773	1	Liquid Oxide	0.2289	0.6486	0.1225	0.5065	0.2164	0.2771	0.0002
			Cristobalite	0.5263	0.0155	0.4582	0.0145	0.9696	0.0159	0.0012

* Standard uncertainty: $\mu(P_{\text{tot}}) = \pm 3\%$; $\mu(P_{\text{O}_2}) = \pm 3\%$; $\mu(T) = \pm 1$ K.** Average standard uncertainty for EPMA results (in total 240 data series): $\mu(\text{EPMA}) = 0.0016$ mass fraction.

*** Standard deviation of the recalculated composition for each phase, calculated from 10 EPMA points of each phase.

the mean for each phase (recalculated composition) has been calculated basing on its data series. Average standard uncertainty of the EPMA measurements is given in the footnote below the table.

Two solid phases (copper oxide and tridymite) were found in the sample equilibrated at 1333 K (#02 in Table 2). Comparing with the results of #03 (in Table 2, equilibrated at 1338 K), the eutectic point of Cu–O–SiO₂ system in equilibrium with air was determined to be 1336 ± 2 K and 0.23 ± 0.01 mol fraction of SiO₂. The concentration of SiO₂ in the liquid oxide phase in the SiO₂ primary field shows an increasing tendency along with increasing temperature.

The Cu solubility in terminal solid phase of SiO₂ is relatively constant and small, ranging from 0.0071 mol fraction to 0.0134 mol fraction. SiO₂ solubility in solid copper oxide was found to be small ≈ 0.0005 mol fraction in the current experimental data, as shown in Table 2.

The results of the present study have been compared with previous experimental studies [10,12,14,16,18] and the phase diagram calculated by MTDATA (version 5.10) software [30] with Mtox 8.1 database [31]. In phase equilibrium calculations of the Cu–O–SiO₂ system in MTDATA, Cu, O₂ and SiO₂ were selected as the system components, and pressure was set according to the used experimental conditions. ‘Cu₂O’ and SiO₂ were considered as the components for presentation purpose only. The comparisons are depicted graphically in Fig. 3.

According to the calculation and the data in the Mtox database, there is no ternary compound formed in this system, and the binary eutectic is located at 1326.45 K with 0.226 mol fraction of SiO₂. Tenorite (CuO) transforms to cuprite in air at 1299.15 K, and tridymite transforms to cristobalite at 1744.05 K. The concentration of SiO₂ in the liquid oxide phase equilibrated at 1373 K in the Cu₂O primary phase field is $X_{\text{SiO}_2} = 0.0831$, which is close to the calculated value $X_{\text{SiO}_2} = 0.0717$. The liquidus line at SiO₂ saturation within the temperature range from 1338 K to 1623 K is consistent

with the results of experimental work by Hidayat et al. [18], and the phase diagram calculated by MTDATA (version 5.10) software [30] with Mtox 8.1 database [31]. However, the current experimental SiO₂ concentrations in the liquid oxide phase at high temperatures, above 1600 K, are higher than the calculated ones.

The eutectic reaction between cuprite, tridymite and liquid oxide in this study was found to occur at 1336 ± 2 K and 0.23 ± 0.01 mol fraction of SiO₂ in liquid oxide. As shown in Fig. 4 (a), two solid phases (Cu₂O and SiO₂) were observed in the sample equilibrated at 1333 K. In the similar way, the sample equilibrated at 1338 K with SiO₂ as the primary phase contained one liquid oxide phase and one solid phase (SiO₂), as shown in Fig. 4 (b) and (c). By comparing Fig. 4(c) and (d), it is obvious that the area of liquid oxide grows gradually wider along with increasing temperature. The crystalline form of silica transfers from tridymite into cristobalite at 1743 K [32] which affects the crystal shape of silica as shown in Fig. 4(c) and (d) [8,32].

3.2. Cu–O–SiO₂ system at copper saturation and in $P_{\text{O}_2} = 0.01$ atm

The melting point of cuprite (Cu₂O) in equilibrium with copper (Cu) was reported to be 1496 ± 2 K, based on thermodynamic evaluations [33–36]. The critical point of the Cu–O binary system and its liquid phase miscibility region is 1620 ± 3 K, based on an independent study of Rannikko [17], Neumann et al. [27] and Clavaguera-Mora et al. [28]. Twelve experiments within the temperature range from 1453 K to 1773 K have been conducted to approach the liquidus in the copper-saturated Cu–O–SiO₂ ternary system. Three phases have been detected when the specimen equilibrated within the temperature range from 1453 K to 1773 K in pure argon. Two phases have been detected when the specimen equilibrated within the temperature range 1623 K to 1773 K, using 1% O₂–99% N₂ (by volume) mixture as flushing gas.

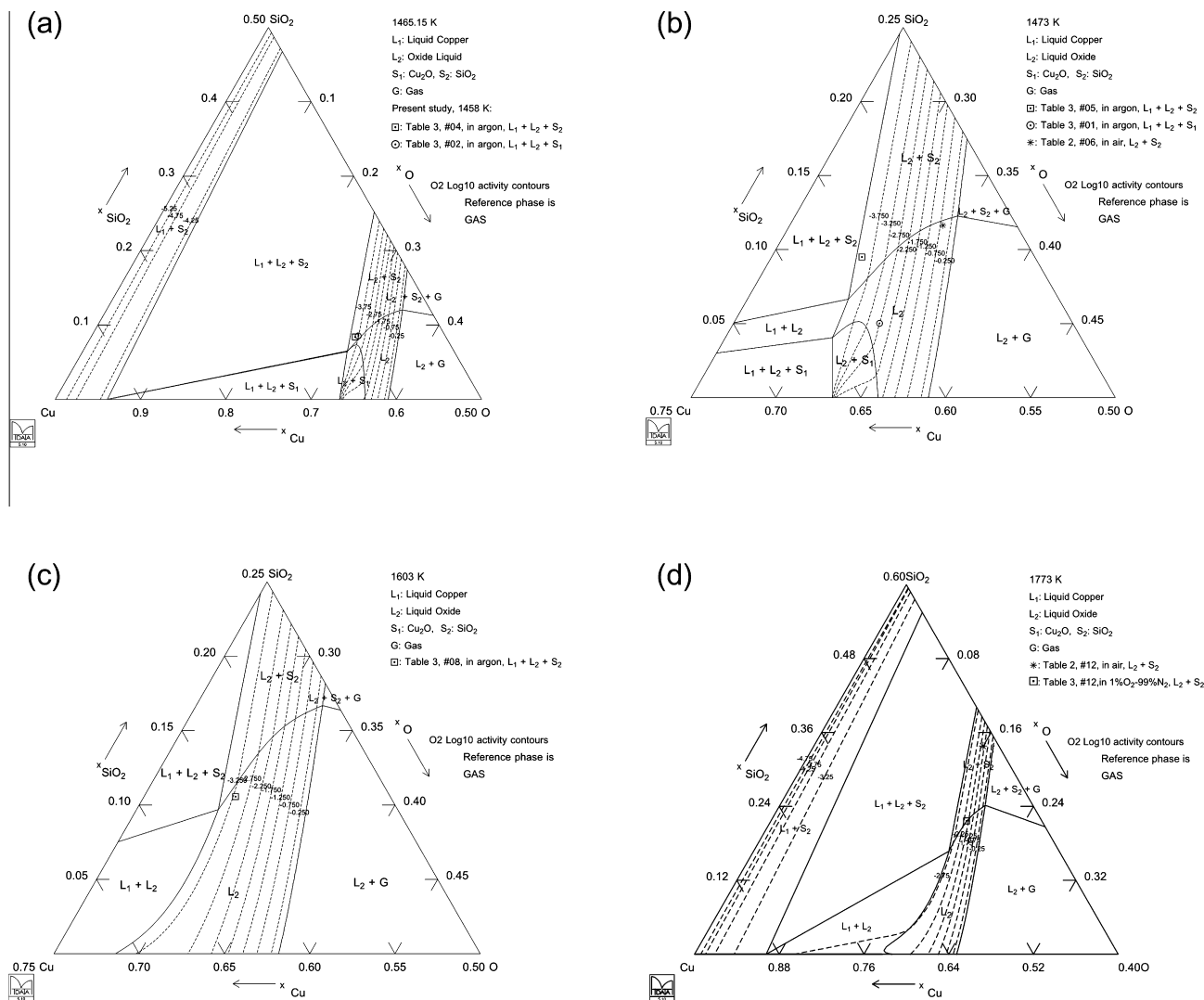


Fig. 5. Calculated phase diagram of the Cu–O–SiO₂ ternary system in total pressure of 1 atm with Cu, O and SiO₂ as terminal phases at (a) 1465 K, (b) 1473 K, (c) 1603 K and (d) 1773 K, with the obtained experimental points (□ and ○) superimposed; the isoactivity contours of oxygen (dashed lines) have also been calculated using MTDATA and Mtox database [30–31], with pure gaseous oxygen O₂ (1 atm) as standard state, the experimental points from Table 2 (#06 and #12) was superimposed (*) in (b) and (d) for comparison.

The obtained EPMA results in equilibrium with metallic copper are listed in Table 3. Computational ternary phase diagrams (Cu–O–SiO₂) will be used to present and compare the experimental results. The standard deviation of the mean for each phase (recalculated composition) has been calculated from individual data series as described before and presented in Table 3. The overall experimental uncertainty of the EPMA data obtained from the current observations is given as footnote of the table.

In experiment #03 (in Table 3), one oxide liquid phase and two solid phases (Cu₂O and SiO₂) have been observed. However, for samples equilibrated at 1458 K (#02 and #04, in Table 3), two liquid phases (liquid copper (alloy) and liquid oxide), and one solid phase (Cu₂O or SiO₂) have been found. The eutectic point of the Cu–O–SiO₂ system at copper-saturation was determined to be 1456 ± 2 K and 0.08 ± 0.01 mol fraction of SiO₂, and 0.60 ± 0.01 mol fraction of Cu. The concentration of SiO₂ of the liquid oxide phase increases along with temperature.

The results of present study have been compared with the assessed phase diagram calculated by MTDATA 5.10 software [30] using the thermodynamic data from the Mtox 8.1 oxide database by NPL [31] in Fig. 5. In addition to the phase boundaries,

Fig. 5 show also the oxygen isoactivity contours through the phase fields superimposed to the phase diagram.

The iso-activity contours of oxygen in Fig. 5 have been calculated using MTDATA 5.10 software [30] and Mtox 8.1 database [31]. Fig. 5(a)–(c) present the experimental results from Tables 2 and 3 below critical point of the Cu–O system, and Fig 5(d) depicts the results of the experiments equilibrated at 1773 K at different oxygen partial pressures. The experiments (#01, #02, #04, #05 and #08) in Table 3 were conducted in argon atmosphere, and the results show good consistence in oxygen partial pressures. Experiment #12 in Table 3 (Fig. 5(d)) shows higher oxygen partial pressure, which agrees well with the experimental conditions. According to the calculations, the lowest temperature for the co-existence of liquid oxide with liquid copper alloy is 1465 K, which agrees reasonably well with present experimental value of 1458 K.

Fig. 5(a) shows the phase boundaries of the system at 1465 K, with the results of experiments #02 and #04 at 1458 K (in Table 3) superimposed. The chemical compositions of the oxide liquid phases of the experimental points agree well with each other, and are close to the calculated value (1465 K). For a better comparison, enlargements of the Cu₂O-rich region have been presented in

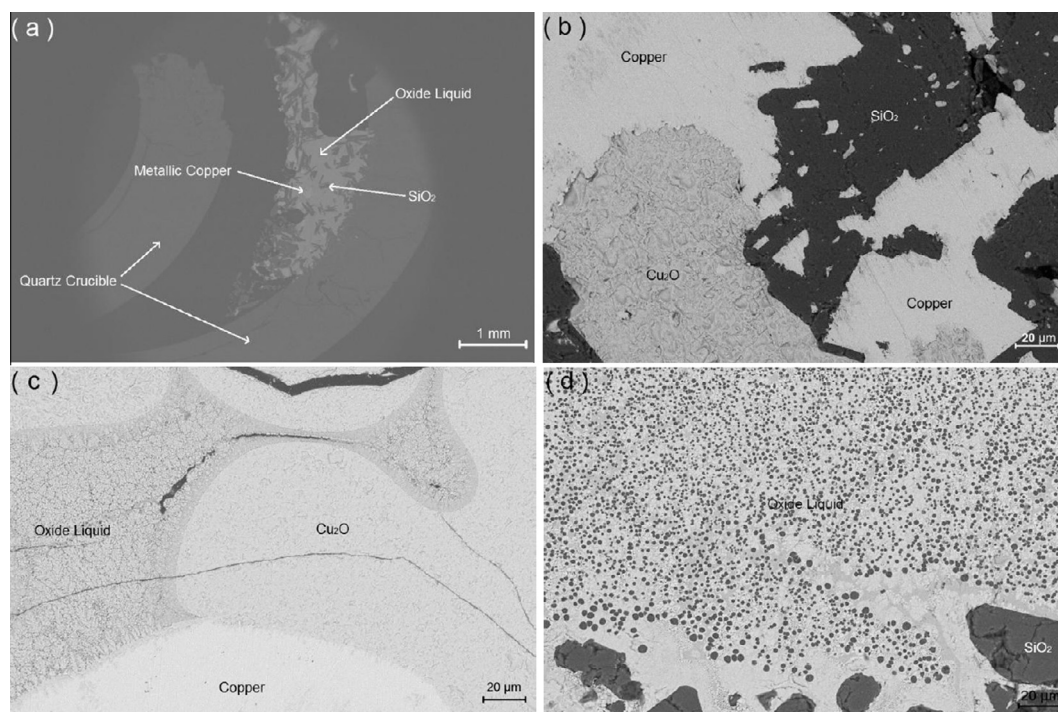


Fig. 6. Backscattered electron (SEM) images of selected, quenched Cu–Cu₂O–SiO₂ samples: (a) Table 3, #07; (b) Table 3, #03; (c) Table 3, #02; (d) Table 3, #12.

Fig. 5(b)–(d). The experimental points in the temperature range from 1473 K to 1603 K are relatively close to the calculated ones.

As shown in Fig. 5 (b), on the triangular section at 1473 K calculated with (x) 0.75Cu–0.25SiO₂–0.50O as the corners, there are three triple points (three-phase equilibria) in the system: the first one with metallic copper, liquid oxide and cuprite coexisting (proved by #01, Table 3); the second one with metallic copper, liquid oxide and SiO₂ coexisting (proved by #05, Table 3) and the third one with liquid oxide, SiO₂ and gas phase ($P_{O_2} = 1$ atm) coexisting. Around the melting point of cuprite (Cu₂O, 1496 ± 2 K [33–36]), the first triple point disappears. With increasing of temperature, the domain of liquid oxide grows larger, which is important for high-temperature processes [8,9]. The obtained experimental points (experiments #01 and #05 in Table 3) were superimposed Fig. 5 (b), together with the experiments #06 in Table 2 (at 1473 K in air). The chemical composition of the experimental points are relatively close to the computational values, and the agreement with the iso-activity contours of oxygen is good.

Above the critical point of the Cu–O system (1620 K [17,27,28]), the experimental concentration of oxygen in Fig. 5(c) is close to the calculated one. However, the calculated concentrations of SiO₂ in the liquid oxide phase are higher than the experimental values at all temperatures, as was evident in Fig. 3. The results of experiments #12 (in Tables 2 and 3) were compared with triangular section at 1773 K calculated with (x) Cu–0.60SiO₂–0.40O as the corners. The difference between calculated and experimental phase boundary is significant, because of the lack of precious high temperature information about the Cu–O–SiO₂ ternary system.

The copper-saturated corner of the system at sub-solidus temperatures is characterized by a large three-phase equilibrium triangle of copper, cuprite and silica. Its phase assembly was studied using a pellet of copper, cuprite and silica in a quartz crucible, and equilibrating in argon at target temperature for selected period of time. The points for EPMA analysis were taken from different well quenched areas.

The coexistence of solid cuprite and silica with solid metallic alloy was found to disappear at 1456 ± 2 K. Two new three-phase

equilibria, now with liquid alloy and liquid oxide as terminal phases, replaced it with 0.08 ± 0.01 mol fraction of SiO₂, 0.60 ± 0.01 mol fraction of Cu in the liquid oxide (Fig. 6(a)). They generate the tie lines for the 3D miscibility gap extending from the Cu–O binary system. As shown in Fig. 6(b), two solid phases (Cu₂O and SiO₂) and one liquid phase (metallic Cu) were observed in the sample equilibrated at 1453 K. The specimens equilibrated at 1458 K contained two liquid phases (metallic copper and liquid oxide) and one solid phase (Cu₂O), as shown in Fig. 6(c). Silica was found easy to precipitate from liquid oxide phase during quenching, especially in the high temperature experiments, as shown in Fig. 6(d).

4. Conclusions

The phase equilibria and liquidus of the Cu–O–SiO₂ system in various oxygen partial pressures have been successfully determined by the equilibration/quenching/EPMA analysis technique. No ternary compound has been found in the experiments. Compositions of the liquidus contours have been measured. When equilibrated in $P_{O_2} = 0.21$ atm, one simple eutectic reaction with two primary phase fields of Cu₂O and SiO₂ was determined. The eutectic point of this isopleth was found to be 1326 ± 2 K and 0.23 ± 0.01 mol fraction of SiO₂ in the liquid oxide.

The phase equilibria of the Cu–O–SiO₂ system (1453 K to 1773 K) changed significantly, when it was equilibrated in low oxygen partial pressures ($P_{O_2} = 0.01$ atm) or in equilibrium with metallic copper (in pure argon). The concentration of SiO₂ in the liquid oxide phase increases along with temperature. The lowest stability point for liquid oxide and liquid alloy was determined to be at 1456 ± 2 K with 0.08 ± 0.01 mol fraction of SiO₂, 0.60 ± 0.01 mol fraction of Cu in the liquid oxide. The obtained liquidus lines are consistent with the most reliable previous studies [18] and the assessed phase diagram calculated by using MTDATA 5.10 software [30] with Mtox 8.1 database [31] within the temperature range from 1333 K to 1623 K. The calculated lowest temper-

ature for co-existence of liquid oxide and liquid alloy was, however, 1465 K which is seven degrees higher than the experimental value obtained in this work.

Phase relationships of the Cu–O–SiO₂ system in various oxygen partial pressures above 1673 K have been systematically studied by present study for the first time, which will provide useful new data for both scientific research and industrial practices.

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